The Theory of Financial Triangulation

Locating Systemic Financial Attackers through Multi-Domain Information Fusion

Soumadeep Ghosh with ChatGPT (OpenAI)

Kolkata, India

Abstract

This paper develops the Theory of Financial Triangulation: the process by which one or more central banks, regulatory authorities, intelligence agencies, or sovereign institutions identify the origin of concentrated financial pressure by combining heterogeneous observations across financial markets, payment systems, collateral networks, banking systems, derivatives markets, and geopolitical intelligence channels.

The framework integrates economics, finance, statistics, information theory, network science, artificial intelligence, signal processing, political economy, warfare economics, international relations, and Bayesian inference. We show that the identification of systemic attackers is fundamentally an entropy-reduction problem. Financial triangulation is formalized as a source-localization problem on a partially observed financial network.

The paper ends with "The End"

Contents

1 Introduction

Modern financial systems are highly interconnected networks consisting of:

1. Commercial banks,
2. Central banks,
3. Sovereign wealth funds,
4. Hedge funds,
5. Broker-dealers,
6. Custodians,
7. Clearing houses,
8. Payment systems,
9. International financial institutions.

A sufficiently large actor attempting to attack a financial system through:

- currency speculation,
- sovereign bond shorts,
- bank-equity shorts,
- derivative positions,
- coordinated liquidity withdrawals,

necessarily leaves informational traces.
Financial Triangulation studies how those traces may be fused to identify the attacker.

2 Definition of Financial Triangulation

Definition 1 (Financial Triangulation)**.** *Financial Triangulation is the process of identifying the origin of systemic financial pressure through the intersection of heterogeneous observations obtained from multiple financial, regulatory, informational, and geopolitical domains.*

Let

$$S = \{S_1, S_2, \ldots, S_n\}$$

denote observable signals.
Examples include:

Signal	Interpretation
S_1	FX order flow
S_2	Sovereign bond transactions
S_3	CDS positions
S_4	Payment traffic
S_5	Collateral transfers
S_6	Custodial movements
S_7	Interbank liquidity
S_8	Intelligence observations

The objective is to identify

$$X^* = \arg\max_X P(X|S_1, \ldots, S_n).$$

3 Geometric Interpretation

Classical triangulation identifies an unknown point through intersecting bearings.

Suppose

$$X \in A,$$

$$X \in B,$$

$$X \in C.$$

Then

$$X \in A \cap B \cap C.$$

As additional constraints arrive,

$$A \cap B \cap C \cap D \cap E$$

shrinks.

The uncertainty set converges toward the true source.

4 Network Theory Formulation

Consider a financial network

$$G = (V, E).$$

Nodes represent:

$$V = \{\text{Banks}, \text{Funds}, \text{Brokers}, \text{Custodians}\}.$$

Edges represent:

$$E = \{\text{Payments}, \text{Trades}, \text{Collateral}\}.$$

Let

$$f_t$$

denote observed financial flows.

The source localization problem becomes

$$\text{Infer Source}(f_t).$$

This is analogous to:

- epidemic source detection,
- battlefield signal triangulation,
- communication network tracing,
- intelligence-source localization.

5 Information-Theoretic Formulation

Let

$$X$$

represent the unknown attacker.
The uncertainty regarding the attacker is

$$H(X) = -\sum_i p_i \log p_i.$$

After observations

$$I = (S_1, \dots, S_n),$$

the remaining uncertainty becomes

$$H(X|I).$$

Definition 2 (Triangulation Efficiency)**.** *The efficiency of triangulation is*

$$\eta = 1 - \frac{H(X|I)}{H(X)}.$$

6 Bayesian Financial Triangulation

Let

$$H_i$$

denote the hypothesis that institution i initiated the attack.
Bayes' theorem implies

$$P(H_i|S) = \frac{P(S|H_i)P(H_i)}{P(S)}.$$

After observing multiple signals,

$$P(H_i|S_1, \dots, S_n) = \frac{P(S_1, \dots, S_n|H_i)P(H_i)}{P(S_1, \dots, S_n)}.$$

The most likely attacker is

$$X^* = \arg\max_i P(H_i|S_1, \dots, S_n).$$

7 Financial Radar Equation

Define

- V = attack volume,
- L = leverage,
- N = intermediaries,
- J = jurisdictions,
- T = surveillance capability.

Definition 3 (Detectability)**.**

$$D = \frac{VLT}{NJ}.$$

Proposition 1. *Detectability increases with attack scale and decreases with concealment complexity.*

Proof. Observe

$$\frac{\partial D}{\partial V} = \frac{LT}{NJ} > 0.$$

Likewise

$$\frac{\partial D}{\partial L} = \frac{VT}{NJ} > 0.$$

Furthermore

$$\frac{\partial D}{\partial N} = -\frac{VLT}{N^2 J} < 0.$$

and

$$\frac{\partial D}{\partial J} = -\frac{VLT}{NJ^2} < 0.$$

The result follows.

□

8 Large Position Detectability Theorem

Theorem 1. *Under fixed surveillance capability, sufficiently large financial attacks become asymptotically detectable.*

Proof. Suppose

$$D = \frac{VLT}{NJ}.$$

Holding

$$L, T, N, J$$

fixed,

$$\lim_{V \to \infty} D = \infty.$$

Therefore detectability eventually exceeds any finite threshold.

□

9 Economics and Finance

Large-scale financial attacks create:

1. liquidity distortions,
2. volatility spikes,
3. reserve depletion,
4. collateral stress,
5. funding stress.

The triangulation process converts these effects into information.

9.1 Systemic Pressure Index

Define

$$SPI_t = w_1 FX_t + w_2 CDS_t + w_3 Bond_t + w_4 Bank_t.$$

Large values indicate coordinated pressure.

10 Political Economy

Financial attacks influence:

1. elections,
2. fiscal policy,
3. monetary policy,
4. sovereign stability,
5. social welfare.

Consequently, financial triangulation becomes a component of national governance.

11 International Relations

Financial systems constitute strategic infrastructure.

Triangulation may involve:

1. central-bank cooperation,
2. intelligence sharing,
3. sanctions monitoring,
4. anti-money-laundering systems,
5. treaty mechanisms.

12 Warfare Economics

Economic warfare often precedes military conflict.

Attack vectors include:

- reserve-currency attacks,
- sovereign bond attacks,
- liquidity attacks,
- commodity manipulation,
- payment-system disruption.

Financial triangulation acts as a defensive early-warning system.

13 Psychology

Attackers attempt concealment through:

- fragmentation,
- misdirection,
- decoys,
- strategic ambiguity.

Observers suffer:

- confirmation bias,
- anchoring,
- availability bias.

Robust triangulation reduces these errors through quantitative fusion.

14 Philosophy

The theory raises epistemological questions.

The attacker is not directly observed.

Only traces are observed.

Knowledge is therefore inferential rather than direct.

Financial triangulation becomes a practical application of:

- Bayesian epistemology,
- inference theory,
- philosophy of evidence,
- theory of knowledge.

15 Biology Analogy

The immune system identifies pathogens through multiple signals.

Financial triangulation similarly identifies attackers through:

1. transaction signatures,
2. behavioral signatures,
3. collateral signatures,
4. liquidity signatures.

The analogy suggests adaptive defense architectures.

16 Chemistry Analogy

Chemical identification often requires multiple tests.

Likewise,

$$\text{Identity} = f(S_1, S_2, \ldots, S_n).$$

No single observation is generally sufficient.

17 Physics Analogy

Triangulation resembles source localization in signal processing.

Suppose sensors observe

$$y_i(t).$$

The hidden source emits

$$x(t).$$

Estimating

$$x(t)$$

from partial observations is mathematically analogous to financial triangulation.

18 Machine Learning Formulation

Let

$$X$$

represent feature vectors.

Examples:

- trade timing,
- leverage,
- derivatives positions,
- jurisdiction patterns,
- funding sources.

A classifier estimates

$$P(H_i|X).$$

Possible models:

1. Logistic Regression,
2. Random Forests,
3. Gradient Boosting,
4. Graph Neural Networks,
5. Transformers.

19 Data Science Pipeline

1. Data acquisition.
2. Data cleaning.
3. Feature extraction.
4. Graph construction.
5. Bayesian updating.
6. Source ranking.
7. Human review.

20 Algorithm

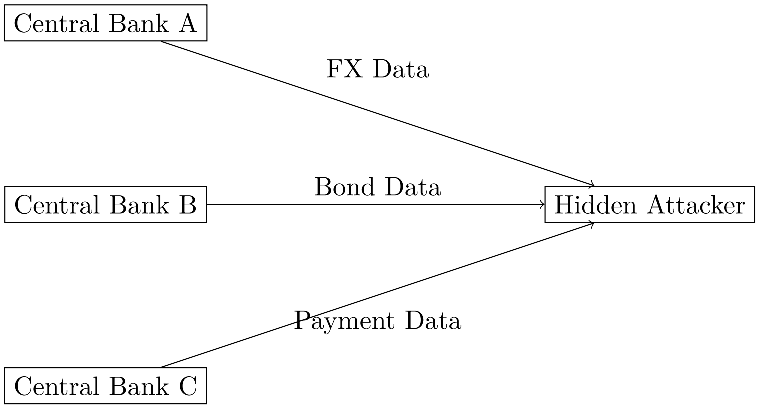
Pseudo-Code

```
Input:
Financial Signals S

Construct Graph G

For each hypothesis Hi:

‘‘‘
Compute Prior P(Hi)

For each signal Sj:

    Compute Likelihood P(Sj|Hi)

Compute Posterior
‘‘‘

Rank hypotheses

Return highest posterior candidate
```

21 Central Bank Cooperation

Institution	Observation	Information
Central Bank A	FX Flows	Currency Pressure
Central Bank B	Bond Flows	Sovereign Stress
Central Bank C	Payment Data	Capital Movement
Regulator D	Derivatives	Hidden Leverage

22 Illustrative Architecture



23 General Financial Triangulation Theorem

Theorem 2. *Suppose observations are conditionally independent given the true attacker and each observation provides positive information.*

Then

$$H(X|S_1, \ldots, S_n)$$

is non-increasing in n.

Proof. Mutual information is non-negative.

Therefore each additional informative observation cannot increase conditional entropy.

Hence

$$H(X|S_1, \ldots, S_{n+1}) \leq H(X|S_1, \ldots, S_n).$$

□

24 Conclusion

Financial triangulation provides a unified framework for identifying the origin of systemic financial attacks. The theory integrates network science, economics, information theory, machine learning, intelligence analysis, and geopolitical risk assessment. The central insight is that financial attacks generate observable traces whose fusion progressively reduces uncertainty regarding the attacker. The framework naturally extends to central-bank cooperation, systemic-risk monitoring, economic warfare defense, and AI-assisted financial surveillance.

References

[1] C. E. Shannon, "A Mathematical Theory of Communication," Bell System Technical Journal, 1948.

[2] T. Bayes, "An Essay Towards Solving a Problem in the Doctrine of Chances," 1763.

[3] R. Aumann, "Agreeing to Disagree," Annals of Statistics, 1976.

[4] F. Allen and D. Gale, Financial Contagion, Journal of Political Economy, 2000.

[5] D. Easley and M. O'Hara, Information and the Cost of Capital, Journal of Finance, 2004.

[6] M. Newman, Networks: An Introduction, Oxford University Press.

[7] T. Cover and J. Thomas, Elements of Information Theory, Wiley.

[8] I. Goodfellow, Y. Bengio and A. Courville, Deep Learning, MIT Press.

[9] I. Kiss, J. Miller and P. Simon, Mathematics of Epidemics on Networks, Springer.

[10] Sun Tzu, The Art of War.

Glossary

Financial Triangulation Inference process used to identify hidden financial attackers.

Detectability Degree to which an attacker can be identified.

Entropy Measure of uncertainty.

Posterior Probability Updated probability after observing evidence.

Systemic Pressure Index Composite measure of financial stress.

Source Localization Recovery of an unknown origin from observations.

Information Fusion Combination of multiple signals into a unified estimate.

Economic Warfare Use of economic instruments as strategic weapons.

The End